

Olympiad Test Paper — Mathematics (Class 7)

Chapter 9: Geometric Twins

Subject	Mathematics (NCERT Class 7)
Chapter	9 — Geometric Twins (Congruence of Triangles)
Total Questions	60 (Multiple Choice, single correct)
Maximum Marks	240
Suggested Time	90 minutes

Marking scheme: +4 for each correct answer · -1 for each wrong answer · 0 if unattempted.

Instructions: Each question has four options (A), (B), (C), (D); exactly one is correct. "Congruent" means same shape AND same size — never confuse it with "similar" (same shape, any size). The classic traps here are: treating AAA or SSA as congruence rules (they are not), forgetting that SAS needs the *included* angle, and answering the wrong angle in an isosceles triangle. Read every figure description carefully. Do not write on this paper — mark answers on your answer sheet.

1. Two figures are said to be congruent when:

- (A) they have the same shape and the same size (B) they have the same area only
(C) they have the same shape only (D) one is an enlargement of the other

2. The symbol used to denote "is congruent to" is:

- (A) $\approx$ (B) $\cong$
(C) $=$ (D) $\sim$

3. A drawing is photocopied at 150% (enlarged). The copy and the original are:

- (A) congruent (B) neither similar nor congruent
(C) similar but not congruent (D) congruent only if the original was a triangle

4. Which set of equal parts does NOT, by itself, guarantee that two triangles are congruent?

- (A) three pairs of equal sides (B) two sides and the included angle
(C) two angles and the included side (D) three pairs of equal angles

5. Two triangles have all three pairs of corresponding sides equal. The criterion that proves them congruent is:

- (A) SAS (B) SSS
(C) ASA (D) RHS

- 6.** In an isosceles triangle the two equal sides each measure x cm. Which other measurement is always equal to its partner across the axis of symmetry?
- (A) the base only (B) the apex angle's two halves only
(C) the two base angles (D) nothing — only the equal sides match
- 7.** Why are two side-lengths NOT enough to copy a triangle exactly, while three are?
- (A) two sides give a triangle that is too small (B) two sides leave the perimeter unknown
(C) two sides always force a right angle
(D) the two given arcs meet at two points, so two different triangles fit
- 8.** In $\triangle ABC$ and $\triangle DEF$, $AB = DE$, $BC = EF$ and $\angle B = \angle E$. By which criterion are they congruent?
- (A) SAS (B) ASA
(C) SSS (D) RHS
- 9.** The "included angle" between two named sides of a triangle is:
- (A) the angle opposite the longer side (B) the largest angle of the triangle
(C) the angle formed where those two sides meet
(D) any angle not touching either side
- 10.** In $\triangle PQR$ and $\triangle XYZ$, $\angle P = \angle X$, $\angle Q = \angle Y$ and $PQ = XY$. The correct congruence statement and criterion is:
- (A) $\triangle PQR \cong \triangle YXZ$ by ASA (B) $\triangle PQR \cong \triangle XYZ$ by SAS
(C) they need not be congruent (AA only) (D) $\triangle PQR \cong \triangle XYZ$ by ASA
- 11.** Each angle of an equilateral triangle measures:
- (A) 60° (B) 90°
(C) 45° (D) 120°
- 12.** An isosceles triangle has apex (vertex) angle 40° . Each base angle measures:
- (A) 40° (B) 70°
(C) 100° (D) 140°
- 13.** In $\triangle ABC$, $\angle B = \angle C = 50^\circ$. Which two sides must be equal?
- (A) $AB = BC$ (B) $BC = AC$
(C) all three are equal (D) $AB = AC$
- 14.** Two right-angled triangles have equal hypotenuses and one pair of equal legs. They are congruent by:
- (A) RHS (B) SSS
(C) SAS (D) ASA
- 15.** A triangle has angles 50° and 60° . A second triangle also has angles 50° and 60° but is drawn twice as large. The two triangles are:

- (A) congruent by ASA (B) similar but not congruent
(C) congruent by AAA (D) congruent by SAS

16. In isosceles $\triangle ABC$, $AB = AC$ and $\angle A = 80^\circ$. The value of $\angle B$ is:

- (A) 80° (B) 40°
(C) 50° (D) 100°

17. To prove the base angles of an isosceles triangle equal, we drop the altitude from the apex and show the two halves are congruent. The criterion used for the two halves is:

- (A) RHS (B) SAS
(C) SSS (D) ASA

18. Two triangles meet the AAA condition (all three angles equal). We can be sure they are:

- (A) congruent (B) the same shape but possibly different size
(C) the same size but possibly different shape (D) right-angled

19. In $\triangle ABC$, $AB = 6$ cm, $AC = 8$ cm, $\angle A = 55^\circ$; in $\triangle DEF$, $DE = 6$ cm, $DF = 8$ cm, $\angle D = 55^\circ$. The triangles are congruent because the equal angle is:

- (A) opposite a side (SSA) (B) the largest angle (AAS)
(C) included between the two equal sides (SAS) (D) a right angle (RHS)

20. $\triangle ABC$ has $AB = 5$ cm, $BC = 7$ cm and $\angle A = 40^\circ$ (the angle at A, NOT between AB and BC). A second triangle has the same three measurements in the same role. We can conclude the two triangles are:

- (A) certainly congruent by SAS (B) certainly congruent by ASA
(C) certainly congruent by SSS
(D) not necessarily congruent — this is the SSA (ambiguous) case

21. If $\triangle ABC \cong \triangle DEF$, $\angle A = 50^\circ$ and $\angle B = 65^\circ$, then $\angle F$ equals:

- (A) 50° (B) 65°
(C) 115° (D) 130°

22. If $\triangle ABC \cong \triangle PQR$, which side of $\triangle PQR$ equals BC?

- (A) PQ (B) PR
(C) QR (D) cannot be decided

23. CPCT stands for "Corresponding Parts of Congruent Triangles (are equal)." It may be used:

- (A) before proving the triangles congruent, to set up the proof
(B) only for the sides, never for the angles (C) only when the triangles are equilateral
(D) only after the triangles have been proved congruent

24. A triangle has two equal sides of 5 cm and an included angle of 60° between them. This triangle is:

- (A) equilateral (B) right-angled
(C) isosceles but never equilateral (D) scalene

25. The minimum number of independent measurements needed to fix a triangle's shape and size is:

- (A) 2 (B) 5
(C) 3 (D) 6

26. In $\triangle ABC$, $\angle A = 70^\circ$ and $\angle B = 70^\circ$. The triangle is:

- (A) equilateral (B) isosceles with $AB = AC$
(C) right-angled (D) isosceles with $AC = BC$

27. Two triangles have sides 4 cm, 6 cm, 9 cm and 4 cm, 6 cm, 9 cm respectively. They are:

- (A) congruent by SSS (B) similar but not congruent
(C) congruent by SAS (D) not necessarily congruent

28. In $\triangle ABC$, $AB = AC$. If $\angle B = (2x + 10)^\circ$ and $\angle C = (3x - 20)^\circ$, then x equals:

- (A) 10 (B) 30
(C) 6 (D) 50

29. A square is folded along one of its diagonals. The two triangular halves are:

- (A) similar but not congruent
(B) congruent only if the square is also a rectangle
(C) never congruent (D) congruent

30. In $\triangle ABC \cong \triangle DEF$, $AB = 7$ cm and $DF = 9$ cm. Given the order of letters, the length of AC is:

- (A) 9 cm (B) 7 cm
(C) cannot be found (D) 16 cm

31. Which statement is FALSE?

- (A) every equilateral triangle is isosceles (B) similar triangles are always congruent
(C) congruent triangles are always similar (D) AAA does not prove congruence

32. In $\triangle PQR$, $PQ = PR$ and $\angle Q = 65^\circ$. Then $\angle P$ equals:

- (A) 65° (B) 25°
(C) 50° (D) 130°

33. Two triangles are drawn: the first with base 5 cm and base angles 50° and 60° ; the second with base 5 cm and base angles 50° and 60° in the same positions. They are congruent by:

- (A) ASA (B) SAS
(C) SSS (D) RHS

34. A right triangle has legs 6 cm and 8 cm. A second right triangle has hypotenuse 10 cm and one leg 6 cm. These two triangles are:

- (A) not congruent
- (B) congruent (the second's other leg is also 8 cm)
- (C) similar only
- (D) congruent only if both are isosceles

35. $\triangle ABC \cong \triangle DEF$ is written with that exact letter order. This tells us $\angle A$ corresponds to:

- (A) $\angle E$
- (B) $\angle F$
- (C) $\angle D$
- (D) any of the three

36. In $\triangle ABC$, AD is the median to BC and $AB = AC$. Comparing $\triangle ABD$ and $\triangle ACD$, they are congruent by:

- (A) ASA
- (B) RHS
- (C) AAA
- (D) SSS

37. The exterior angle at the apex of an isosceles triangle is 100° . Each base angle is:

- (A) 100°
- (B) 50°
- (C) 80°
- (D) 40°

38. Two triangles cannot be congruent if they have:

- (A) equal perimeters and an SSS match
- (B) the same three angles
- (C) different areas
- (D) two equal sides and the included angle equal

39. In $\triangle LMN$ and $\triangle RST$, $LM = RS$, $\angle L = \angle R$ and $\angle M = \angle S$. The congruence criterion is:

- (A) SAS
- (B) SSS
- (C) SSA
- (D) ASA

40. An isosceles triangle has a base angle of 75° . Its apex angle is:

- (A) 30°
- (B) 75°
- (C) 105°
- (D) 15°

41. In $\triangle ABC$, $\angle A = 90^\circ$, $AB = AC$. Then $\angle B$ equals:

- (A) 90°
- (B) 60°
- (C) 45°
- (D) 30°

42. Which of these guarantees congruence of two RIGHT triangles using exactly three matching parts that include the right angle?

- (A) all three angles equal
- (B) one acute angle equal (and the right angle), but no side given
- (C) the perimeters equal
- (D) the two pairs of legs equal (and the right angle between them)

43. A triangle is fixed by its three sides 3 cm, 4 cm, 5 cm. How many genuinely different (non-congruent) triangles have exactly these three side-lengths?

- (A) exactly one
- (B) none
- (C) exactly two
- (D) infinitely many

- 44.** In $\triangle ABC$, $AB = AC$ and $\angle A = 100^\circ$. The largest angle of the triangle is:
 (A) each base angle at 40° (B) $\angle A$ at 100°
 (C) $\angle B$ at 80° (D) all equal
- 45.** $\triangle ABC$ has $AB = AC$. The bisector of $\angle A$ meets BC at D . Then BD compared with DC is:
 (A) $BD > DC$ (B) $BD < DC$
 (C) cannot be determined (D) $BD = DC$
- 46.** Two triangles have two pairs of equal sides and a pair of equal angles, but the equal angle is NOT between the equal sides. This information:
 (A) does not guarantee congruence (SSA, ambiguous)
 (B) always proves congruence by SAS (C) always proves congruence by SSS
 (D) proves the triangles are equilateral
- 47.** In $\triangle ABC$, the perpendicular bisector of side BC passes through A . This means:
 (A) BC is the longest side (B) $AB = AC$, so the triangle is isosceles
 (C) $\angle A = 90^\circ$ (D) the triangle is scalene
- 48.** $\triangle ABC \cong \triangle PQR$. The perimeter of $\triangle ABC$ is 24 cm. The perimeter of $\triangle PQR$ is:
 (A) 12 cm (B) 48 cm
 (C) 24 cm (D) cannot be found
- 49.** In $\triangle ABC$, $\angle A = 40^\circ$ and $AB = AC$. Then $\angle C$ of $\triangle ABC$ is:
 (A) 70° (B) 40°
 (C) 100° (D) 50°
- 50.** Which pair of conditions is enough to construct a UNIQUE triangle?
 (A) three angles (B) two angles and the included side
 (C) two sides only (D) one side and one angle
- 51.** The angles of a triangle are in the ratio 1 : 1 : 2. Such a triangle is:
 (A) equilateral (B) scalene
 (C) isosceles and right-angled (D) impossible
- 52.** In $\triangle DEF$, $DE = DF$ and $\angle D = 2\angle E$. Then $\angle E$ equals:
 (A) 36° (B) 60°
 (C) 90° (D) 45°
- 53.** A triangle has sides 7 cm, 7 cm, 7 cm. Each of its angles is:
 (A) 45° (B) 60°
 (C) 50° (D) 70°
- 54.** $\triangle ABC \cong \triangle DEF$ by SAS, using $AB = DE$, $\angle A = \angle D$ and $AC = DF$. The angle $\angle A$ is the included angle between sides:

(A) AB and BC

(B) BC and AC

(C) AB and AC

(D) the hypotenuse and a leg

55. Two triangles $\triangle ABC$ and $\triangle ABD$ share the common base AB, lie on opposite sides of it, with $AC = AD$ and $BC = BD$. The criterion proving $\triangle ABC \cong \triangle ABD$ is:

(A) SAS

(B) ASA

(C) RHS

(D) SSS

56. In isosceles $\triangle ABC$, $AB = AC$, so $\angle B = \angle C$. If $\angle B = (x + 20)^\circ$ and the apex $\angle A = (2x)^\circ$, then $\angle A$ equals:

(A) 70°

(B) 35°

(C) 50°

(D) 130°

57. A child claims: "If two triangles have the same area, they are congruent." This claim is:

(A) always true

(B) true only for right triangles

(C) false — equal area does not force equal shape and size

(D) true only for equilateral triangles

58. In $\triangle ABC$, $AB = AC = 13$ cm and $BC = 10$ cm. The altitude from A to BC has foot M. The length BM is:

(A) 6.5 cm

(B) 12 cm

(C) 10 cm

(D) 5 cm

59. $\triangle ABC \cong \triangle ACB$ (note the letter order) for a single triangle ABC. This self-congruence forces:

(A) $AB = AC$, i.e. the triangle is isosceles

(B) $\angle A = 90^\circ$

(C) the triangle is scalene

(D) the triangle is equilateral

60. A triangle has angles in the ratio $2 : 2 : 5$. The triangle is:

(A) equilateral

(B) isosceles

(C) right-angled

(D) scalene

End of question paper. Maximum marks: 240. Marking: +4 correct, -1 wrong, 0 unattempted.

Solutions — Mathematics (Class 7), Chapter 9: Geometric Twins

Marking: +4 correct, –1 wrong, 0 unattempted. Maximum marks **240**.

Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	A	11	A	21	B	31	B	41	C	51	C
2	B	12	B	22	C	32	C	42	D	52	D
3	C	13	D	23	D	33	A	43	A	53	B
4	D	14	A	24	A	34	B	44	B	54	C
5	B	15	B	25	C	35	C	45	D	55	D
6	C	16	C	26	D	36	D	46	A	56	A
7	D	17	A	27	A	37	B	47	B	57	C
8	A	18	B	28	B	38	C	48	C	58	D
9	C	19	C	29	D	39	D	49	A	59	A
10	D	20	D	30	A	40	A	50	B	60	B

Answer-key distribution: A = 15, B = 15, C = 15, D = 15.

Worked Solutions

- (A)** Congruent means one figure can be placed exactly on the other — that requires the **same shape AND the same size**. "Same area only" (a 2×6 and a 3×4 rectangle) or "same shape only" (an enlargement) fail this.
- (B)** $\cong$ is the congruence symbol. $\sim$ means "similar", $\approx$ means "approximately", $=$ is ordinary equality.
- (C)** A 150% copy keeps the shape but multiplies every length by 1.5, so it is bigger. Same shape, different size = **similar but not congruent**. (Congruent needs equal size.)
- (D)** SSS, SAS and ASA each pin a triangle down completely. **AAA (three equal angles)** fixes only the shape — the triangles can be different sizes (similar), so it does not guarantee congruence.
- (B)** Three pairs of equal corresponding sides is exactly the **SSS** criterion.
- (C)** Fold an isosceles triangle along its axis of symmetry (the altitude from the apex): the two halves coincide, so the **two base angles** are forced equal. The equal sides face

equal angles.

7. (D) Drawing a base and swinging two arcs from its ends, **the arcs meet at two points** (one above, one below the base), giving two different triangles. A third side removes the ambiguity, so SSS fixes the triangle.

8. (A) $AB = DE$ and $BC = EF$ are two sides, and $\angle B$ is the angle **between** them (the included angle). Two sides + included angle = **SAS**.

9. (C) The included angle of two named sides is **the angle where those two sides meet** (their common vertex). This is the key word in SAS.

10. (D) Two angles ($\angle P = \angle X$, $\angle Q = \angle Y$) with the **included** side $PQ = XY$ give **ASA**, and the vertices already correspond in order $P \leftrightarrow X$, $Q \leftrightarrow Y$, $R \leftrightarrow Z$, so $\triangle PQR \cong \triangle XYZ$. (AA alone would only give similarity, but here a side is also equal.)

11. (A) All three sides equal $\Rightarrow$ all three angles equal; they sum to 180° , so each is $180 \div 3 = 60^\circ$. (Not 90° and not $360 \div 3$.)

12. (B) Apex 40° , so the two equal base angles share $180 - 40 = 140^\circ$, giving $140 \div 2 = 70^\circ$ each. (40° is the apex itself — the trap.)

13. (D) Equal angles are faced by equal sides. $\angle B$ and $\angle C$ are equal, so the sides opposite them — AC (opposite B) and AB (opposite C) — are equal: **AB = AC**.

14. (A) A right angle, equal hypotenuses and one equal leg is precisely the **RHS** criterion. (SSS would need the third side proved equal first; RHS gives it directly.)

15. (B) Equal angles fix only the shape; "twice as large" means a different size. So the triangles are **similar but not congruent** — AAA is not a congruence rule.

16. (C) $AB = AC \Rightarrow \angle B = \angle C$. With $\angle A = 80^\circ$, $2\angle B = 180 - 80 = 100$, so $\angle B = 50^\circ$. (80° is $\angle A$; 100° is the sum of the base angles.)

17. (A) The altitude makes $\angle ADB = \angle ADC = 90^\circ$ (right angle), $AB = AC$ (hypotenuses), AD common (a leg) $\rightarrow$ **RHS**. The matching base angles then equal by CPCT.

18. (B) AAA guarantees only **the same shape but possibly different size** (similar). It is not a congruence criterion.

19. (C) The equal angle $\angle A$ sits **between the two equal sides** AB and AC, so this is the included angle — **SAS**. (If the angle were opposite a side it would be the unreliable SSA case.)

20. (D) Here $\angle A$ is at vertex A while the given sides are AB and BC; $\angle A$ is NOT between them — it is a **non-included angle (SSA)**, which can fit two different triangles. So

congruence is **not guaranteed**.

21. (B) In $\triangle ABC$, $\angle C = 180 - 50 - 65 = 65^\circ$. Congruence $\triangle ABC \cong \triangle DEF$ makes $\angle F$ correspond to $\angle C$, so $\angle F = 65^\circ$.

22. (C) Under $\triangle ABC \cong \triangle PQR$ the correspondence is $A \leftrightarrow P$, $B \leftrightarrow Q$, $C \leftrightarrow R$. Side BC (between B and C) corresponds to **QR** (between Q and R).

23. (D) CPCT lets you conclude corresponding parts are equal — but **only after** the two triangles have been shown congruent. It is a consequence of congruence, never a step to prove it.

24. (A) Two equal sides with a 60° included angle: the third side, by the isosceles property the base angles are equal, and equal base angles with a 60° apex force all angles to 60° , so all three sides become equal — the triangle is **equilateral**. (A 5-5 triangle with a 60° apex is the classic equilateral.)

25. (C) A triangle has six measurements, but only **3** well-chosen independent ones (e.g. SSS, SAS or ASA) are needed to fix it completely. Two are too few; six is unnecessary.

26. (D) $\angle A = \angle B = 70^\circ$, so the sides facing them are equal: side opposite A is BC, side opposite B is AC, hence **AC = BC** (isosceles). (Watch which sides face the equal angles.)

27. (A) All three pairs of sides equal (4, 6, 9 each) $\Rightarrow$ congruent by **SSS**. Equal corresponding sides means same size too, not merely similar.

28. (B) $AB = AC \Rightarrow \angle B = \angle C$: $2x + 10 = 3x - 20 \rightarrow 30 = x$, so $x = 30$. (Check: $\angle B = 70^\circ = \angle C$.)

29. (D) A diagonal splits a square into two triangles with the two sides of the square equal, the included right angle equal, and the diagonal shared — by SAS (or SSS) they are **congruent**. Folding maps one exactly onto the other.

30. (A) $\triangle ABC \cong \triangle DEF$ gives $A \leftrightarrow D$, $B \leftrightarrow E$, $C \leftrightarrow F$. Side AC corresponds to DF, so $AC = DF = 9$ cm. ($AB = DE = 7$ cm is a different pair — the trap.)

31. (B) Congruent $\Rightarrow$ similar (true), every equilateral triangle is isosceles (true), AAA fails to prove congruence (true). The **false** statement is "**similar triangles are always congruent**" — similar triangles can differ in size.

32. (C) $PQ = PR \Rightarrow \angle Q = \angle R = 65^\circ$. $\angle P = 180 - 65 - 65 = 50^\circ$. (65° is a base angle, not the apex.)

33. (A) Base 5 cm with the two base angles (50° and 60°) equal in the same positions: two angles + the included side $\Rightarrow$ **ASA**.

34. (B) First triangle: legs 6, 8 $\Rightarrow$ hypotenuse 10. Second: hypotenuse 10, one leg 6 $\Rightarrow$ other leg $\sqrt{(100 - 36)} = 8$. Right angle + equal hypotenuse + equal leg (RHS) makes them **congruent**, with the missing leg also 8 cm.

35. (C) The letter order in $\triangle ABC \cong \triangle DEF$ pairs the vertices in place: $A \leftrightarrow D$. So $\angle A$ corresponds to $\angle D$.

36. (D) AD is a median $\Rightarrow BD = DC$; $AB = AC$ (given); AD common. Three pairs of equal sides $\Rightarrow$ **SSS**.

37. (B) Exterior angle 100° at the apex $\Rightarrow$ interior apex angle $= 180 - 100 = 80^\circ$. Base angles share $180 - 80 = 100^\circ$, so each $= 50^\circ$. (Using 100° directly as the interior apex is the trap.)

38. (C) Congruent triangles must be identical in size, hence equal area — so two triangles with **different areas** can never be congruent. (Equal perimeter + SSS, equal angles, or SAS are all consistent with congruence.)

39. (D) $\angle L = \angle R$, $\angle M = \angle S$ are two angles, and the side $LM = RS$ lies **between** $\angle L$ and $\angle M$ — included side. Two angles + included side $\Rightarrow$ **ASA**.

40. (A) Base angle $75^\circ \Rightarrow$ both base angles 75° , summing to 150° . Apex $= 180 - 150 = 30^\circ$.

41. (C) $\angle A = 90^\circ$, $AB = AC \Rightarrow \angle B = \angle C$; they share $180 - 90 = 90^\circ$, so each $= 45^\circ$.

42. (D) With the right angle counted as one of the parts, **two pairs of equal legs (with the included right angle)** gives SAS congruence using exactly the right angle + two sides. ("All three angles" is mere AAA — shape only; a right angle + one acute angle with no side given is still only angles (AAA), fixing shape not size; and equal perimeters alone prove nothing.)

43. (A) Three fixed side-lengths leave **exactly one** triangle up to congruence (SSS pins it completely). Any two such triangles can be laid exactly on top of each other.

44. (B) $AB = AC \Rightarrow \angle B = \angle C = (180 - 100)/2 = 40^\circ$. The biggest angle is the apex, $\angle A$ at 100° (each base angle is only 40°).

45. (D) In an isosceles triangle the bisector of the apex angle is also the median and altitude to the base, so it bisects BC: **$BD = DC$** . ($\triangle ABD \cong \triangle ACD$ by SAS, then CPCT.)

46. (A) Two equal sides and an equal angle that is **not** between them is the SSA pattern, which can fit two different triangles. So it **does not guarantee congruence**.

47. (B) Every point on the perpendicular bisector of BC is equidistant from B and C; since A lies on it, $AB = AC$, so the triangle is **isosceles**. (It says nothing about a right angle or the longest side.)

48. (C) Congruent triangles are identical copies, so all corresponding sides are equal and the perimeters match: **24 cm**.

49. (A) $AB = AC \Rightarrow \angle B = \angle C$, and $\angle B + \angle C = 180 - 40 = 140^\circ$, so $\angle C = 70^\circ$. (40° is $\angle A$, not a base angle.)

50. (B) Two angles and the included side (ASA) fix a unique triangle. Three angles give only shape (AAA), two sides alone are ambiguous, and a single side-and-angle is far too little.

51. (C) $1 : 1 : 2$ over 180° gives angles $45^\circ, 45^\circ, 90^\circ$: two equal angles (isosceles) and a right angle — **isosceles and right-angled**.

52. (D) $DE = DF \Rightarrow \angle E = \angle F$. With $\angle D = 2\angle E$: $\angle D + \angle E + \angle F = 2\angle E + \angle E + \angle E = 4\angle E = 180 \rightarrow \angle E = 45^\circ$.

53. (B) All sides equal (7, 7, 7) $\Rightarrow$ equilateral $\Rightarrow$ every angle = **60°** .

54. (C) SAS uses two sides and the angle between them. The named sides are AB and AC, both meeting at A, so $\angle A$ is included between **AB and AC**.

55. (D) $AC = AD, BC = BD$ (given) and AB is the common base — three pairs of equal sides $\Rightarrow$ **SSS**.

56. (A) $AB = AC \Rightarrow \angle B = \angle C = (x + 20)^\circ$; with apex $\angle A = (2x)^\circ$: $2x + (x + 20) + (x + 20) = 180 \rightarrow 4x + 40 = 180 \rightarrow x = 35$. So $\angle A = 2x = 70^\circ$. (Check: base angles 55° each, $70 + 55 + 55 = 180$. ✓)

57. (C) Equal area does NOT force equal shape and size — a tall thin triangle and a short wide one can share an area. So the claim is **false**.

58. (D) $AB = AC$, so the altitude from apex A to base BC also bisects BC (isosceles property): M is the midpoint, giving $BM = BC/2 = 10/2 = 5$ cm. (6.5 wrongly halves a slant side; 12 is the altitude length $\sqrt{13^2 - 5^2}$.)

59. (A) $\triangle ABC \cong \triangle ACB$ pairs $A \leftrightarrow A, B \leftrightarrow C, C \leftrightarrow B$, so AB corresponds to AC $\Rightarrow$ **AB = AC**: the triangle is isosceles. (This need not make it equilateral or right-angled.)

60. (B) $2 : 2 : 5$ over 180° (9 parts of 20°) gives $40^\circ, 40^\circ, 100^\circ$: two equal angles $\Rightarrow$ **isosceles** (and obtuse, but isosceles is the classification asked).